VOICEOPS | CAPSTONE REPORT

ZentroMart customer-support voice agent | Final verified snapshot | 25 October 2026

System architecture and workflow

VoiceOps combines policy/catalog retrieval, deterministic business rules, conversational state, escalation tools, and a speech pipeline. The dataset is supplied separately and is not packaged.

Stage	Method	Verified result
1 - Prompt-only	52 normal-text questions; system prompt without retrieved documents or tools.	23.1% content; 88.5% decision accuracy; 0.000 retrieval recall.
2 - RAG	Policy/catalog retrieval with dated evidence and answer generation.	96.2% content; 96.2% decision accuracy; 0.819 mean recall.
3 - Conversations	96 turns across 23 conversations with identity checks and escalation tools.	99.0% checks/content; 100.0% decision/tool-call accuracy; 22/23 fully correct.
4 - Voice	Gnani STT -> agent -> Gnani TTS; synthetic audio, offline rules agent.	3 STT + 3 TTS calls; 0 errors; human-request turn created a ticket.

Retrieval and deterministic logic

Knowledge-base sections retain evidence IDs, status, and effective dates; catalog rows are searchable alongside policy text. Retrieval ranks relevant evidence, while code handles exact policy dates, order verification, return deadlines, and ticket success checks. Responses record the retrieved evidence IDs and tool calls. The LLM is not the source of truth for deterministic order and policy calculations.

The Step 2 RAG and Step 1 prompt-only figures are historical saved baselines. The later offline rules-only scores below are a separate evaluation and are not live LLM results. No new LLM calls were made for this finalization.

Current quality and validation

Fresh local checks and the previously measured offline evaluations are reported separately from the historical Step 1/Step 2 baselines.

Regression and dataset checks

Check	Observed result
Python compile	PASS; exit code 0.
Full pytest	47 passed; 0 failed.
Dataset validator	PASS; 9 CSVs, 21 KB documents, 1,000 orders, 1,783 items, 244 logs, 52 questions, 23 conversations.

Rules-only offline evaluation

Evaluation	Cases	Content/checks	Decision	Mean retrieval recall	Tool calls
Normal-text questions	52	100.0% content	100.0%	0.989	-
ASR-text questions	52	100.0% content	100.0%	0.989	-
Conversations	96 turns / 23 conversations	100.0% checks/content	100.0%	-	100.0%; 23/23 complete

Interpretation and limitations

The offline runs use deterministic rules; their scores do not establish live LLM-enabled performance. The full 47-test result was verified in the full workspace. The clean package excludes the private hidden-case evaluation file and the separate dataset. Browser microphone use, personal recordings, and live LLM behavior were not tested in this finalization.

Speech integration and reproducibility

The refreshed video alternates each original synthetic customer-input WAV with the corresponding recorded synthesized agent reply WAV. It contains six ordered audio segments (three customer turns and three assistant replies), with readable transcript cards and evidence IDs. Total duration is 33.8 seconds; no new Gnani calls were made to create it.

Three-turn demo content

Turn	Customer input	Agent response / evidence
1 - answer	What is the return window for clothing?	Explains the delivery-date calendar window. Evidence: RET-2, RET-9, RET-3, DWM-4.
2 - answer	Does ZentroMart Plus waive the return pickup fee?	Explains the Rs 49 fee and waiver. Evidence: RET-2, RET-6, PLS-2, FAQH-4.
3 - escalate	Please connect me to a human agent.	Ticket created and handoff confirmed. Evidence: FAQ-4, DWM-5, ESC-1, DWM-3.

Install and run

Use Python 3.9+, create a virtual environment, install requirements.txt, copy the blank .env.example to .env, and set DATASET_DIR and GNANI_API_KEY locally. Optional LLM settings are LLM_PROVIDER, LLM_API_KEY, LLM_MODEL, and LLM_BASE_URL; provider-specific key names are also supported. Never commit credentials.

Normal LLM-capable mode (uses Gnani speech and a configured LLM):

```
python run.py --audio-dir "samples\incoming" --output-dir "output" --language-code en-IN
```

Offline rules-only mode (no LLM calls; real audio still uses Gnani STT/TTS):

```
python run.py --audio-dir "samples\incoming" --output-dir "output_offline" --language-code en-IN --offline
```

The runner writes reply text, reply WAV, per-input JSON, results.csv, and results.jsonl. Use a fresh output directory; existing result files are not overwritten.

Public repository: https://github.com/chukka-venugopalam/voiceops